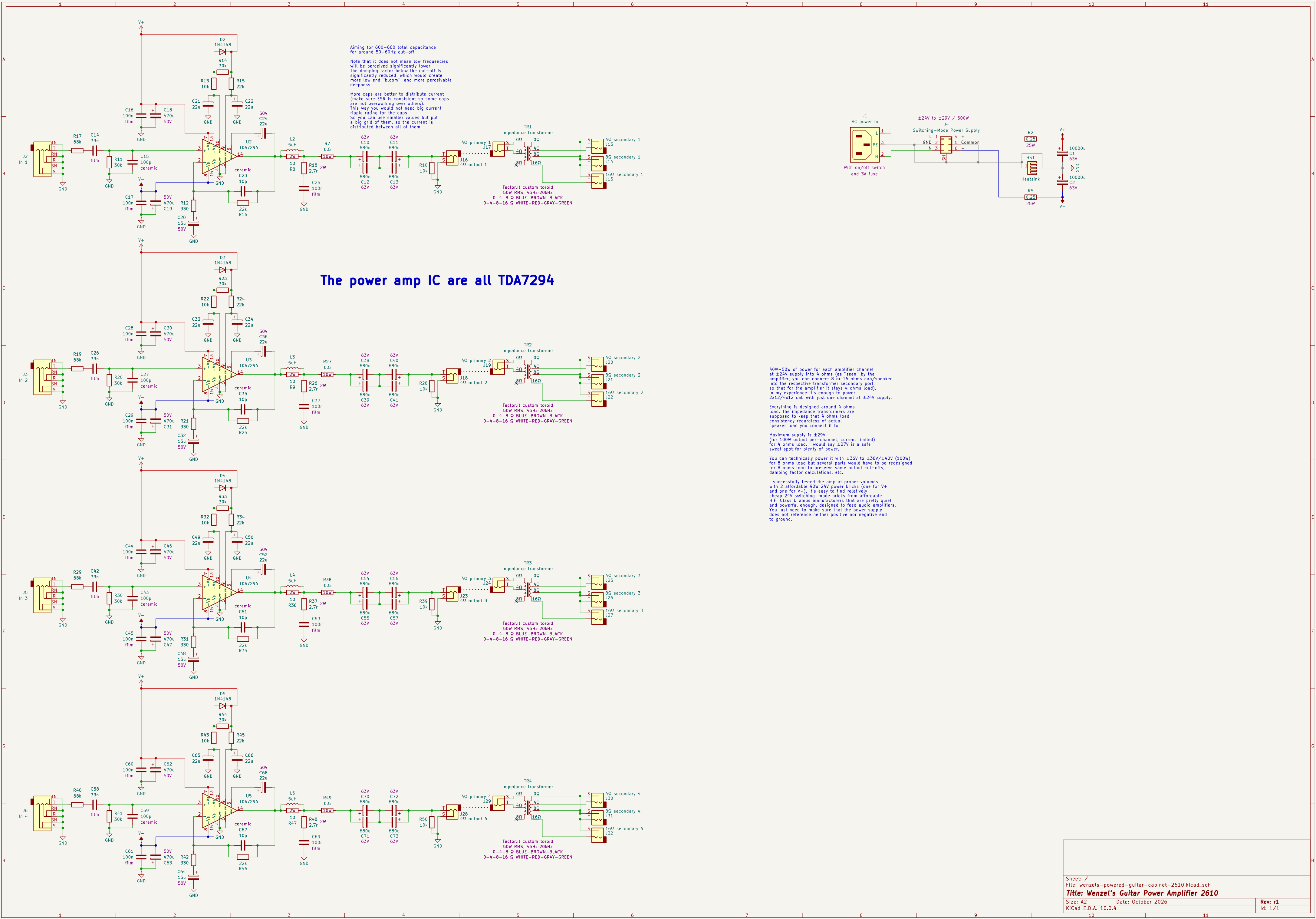




The power amp IC are all TDA7294

Aiming for 600-680 total capacitance for around 50-60Hz cut-off.
Note that it does not mean low frequencies will be perceived significantly lower. The damping factor below the cut-off is significantly reduced, which would create more low end "bloom", and more perceivable deepness.

More caps are better to distribute current (make sure ESR is consistent so some caps are not overworking over others). This way you would not need big current ripple rating for the caps. So you can use smaller values but put a big grid of them, so the current is distributed between all of them.

Tector.it custom toroid
50W RMS, 45Hz-20kHz
0-4-8 Q BLUE-BROWN-BLACK
0-4-8-16 Q WHITE-RED-GRAY-GREEN

Tector.it custom toroid
50W RMS, 45Hz-20kHz
0-4-8 Q BLUE-BROWN-BLACK
0-4-8-16 Q WHITE-RED-GRAY-GREEN

Tector.it custom toroid
50W RMS, 45Hz-20kHz
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Tector.it custom toroid
50W RMS, 45Hz-20kHz
0-4-8 Q BLUE-BROWN-BLACK
0-4-8-16 Q WHITE-RED-GRAY-GREEN

40W-50W of power for each amplifier channel at $\pm 24V$ supply into 4 ohms (as "seen" by the amplifier, you can connect 8 or 16 ohms cab/speaker into the respective transformer secondary port, so that for the amplifier it stays 4 ohms load). In my experience it's enough to power 2x12/4x12 cab with just one channel at $\pm 24V$ supply.

Everything is designed around 4 ohms load. The impedance transformers are supposed to keep that 4 ohms load consistency regardless of actual speaker load you connect it to.

Maximum supply is $\pm 29V$ (for 100W output per-channel, current limited) for 4 ohms load. I would say $\pm 27V$ is a safe sweet spot for plenty of power.

You can technically power it with $\pm 36V$ to $\pm 38V/\pm 40V$ (100W) for 8 ohms load but several parts would have to be redesigned for 8 ohms load to preserve same output cut-offs, damping factor calculations, etc.

I successfully tested the amp at proper volumes with 2 affordable 90W 24V power bricks (one for V+ and one for V-). It's easy to find relatively cheap 24V switching-mode bricks from affordable HIFI Class D amps manufacturers that are pretty quiet and powerful enough, designed to feed audio amplifiers. You just need to make sure that the power supply does not reference neither positive nor negative end to ground.